AIR QUALITY DESCRIPTIVE ANALYSIS

Group 09

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Abstract

This study conducts a descriptive analysis of air quality patterns using 5,000 observations across multiple regions. The analysis examines environmental factors, pollutant concentrations, and demographic characteristics in determining air quality levels (Good, Moderate, Poor, Hazardous). Key findings reveal significant associations between all variables and air quality levels, with severe multicollinearity identified between PM2.5 and PM10 (r = 0.99). Principal Component Analysis reduced dimensionality from 9 to 7 components while retaining 97% variance. Results indicate that hazardous air quality is associated with higher temperatures, elevated pollutant concentrations, closer proximity to industrial areas, and higher population density. The analysis provides insights into air quality determinants and establishes a foundation for environmental policy development and predictive modeling.

Introduction

Air pollution has become one of the most critical environmental challenges faced by both developed and developing nations in the 21st century. The degradation of air quality not only affects ecological balance but also has severe implications for human health and societal wellbeing. As urbanization intensifies and industrial activities expand, understanding the drivers behind varying air quality levels is essential for effective environmental management and policymaking. This dataset provides a comprehensive view of environmental and demographic factors influencing air quality across multiple regions. With 5,000 observations, it includes variables such as pollutant concentrations (PM2.5, PM10, NO_2, SO_2, CO), meteorological indicators (temperature and humidity), and contextual factors (population density and proximity to industrial areas). The target variable, Air Quality Level, is categorized into four levels: Good, Moderate, Poor, and Hazardous, each representing a different degree of pollution severity and public health concern. This study aims to perform a detailed exploratory data analysis to uncover patterns, relationships, and potential drivers of air quality variability.

Description of the Question

With growing global concern over environmental degradation, especially air pollution, it is important to identify the underlying factors that contribute to poor air quality. Regions with dense populations and close proximity to industrial zones may experience different pollution profiles compared to rural or isolated areas. Likewise, meteorological conditions such as temperature and humidity can influence the dispersion and concentration of airborne pollutants. Hence, this report seeks to answer the following key questions: • What are the major environmental and demographic factors associated with poor or hazardous air quality? • How are different pollutant levels (PM2.5, PM10, NO_2, SO_2, CO) distributed across the four air quality categories? • Do certain regions show a higher risk of falling into the “Hazardous” air quality category based on their population density or proximity to industrial areas? Ultimately, the analysis aims to identify influential features and patterns that can guide environmental monitoring, inform policymaking, and lay the groundwork for predictive modeling of air quality based on observed factors.

Description of the Dataset

Overview

This dataset contains air quality measurements and environmental factors for classifying air quality levels across different regions.

Variables

Environmental Factors

- Temperature (°C): Average regional temperature
- Humidity (%): Relative humidity levels
- Proximity to Industrial Areas (km): Distance to nearest industrial zone
- Population Density (people/km²): Regional population per square kilometer

Pollution Concentrations

- PM2.5 Concentration (µg/m³): Fine particulate matter levels
- PM10 Concentration (µg/m³): Coarse particulate matter levels
- NO2 Concentration (ppb): Nitrogen dioxide levels
- SO2 Concentration (ppb): Sulfur dioxide levels
- CO Concentration (ppm): Carbon monoxide levels

Target Variable

Air Quality Levels with four categories:

- Good: Clean air with low pollution
- Moderate: Acceptable air quality with some pollutants
- Poor: Noticeable pollution affecting sensitive groups
- Hazardous: Highly polluted air posing serious health risks

Data Pre-processing

- Missing Values: No missing values were found in the dataset.
- Duplicates: No duplicate records were found in the dataset.
- Data entry errors: No Data entry error.

Univariate Analysis

Distribution of the target variable

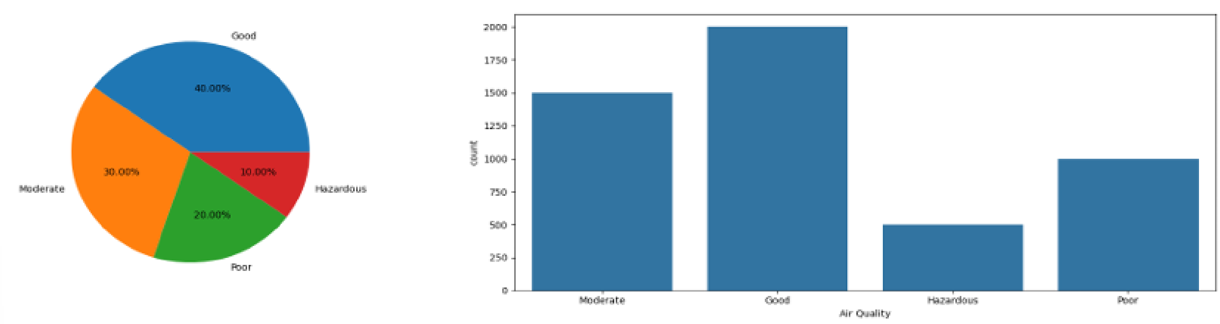
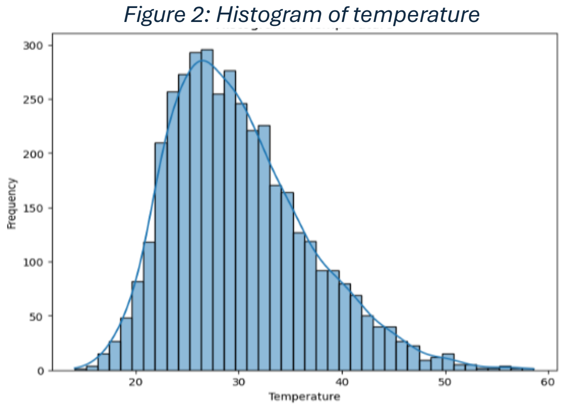


Figure 1: Distribution of Target variable

The distribution of the target variable, air quality category, shows a moderate class imbalance. The majority of instances fall into the "Good" (40%) and "Moderate" (30%) categories, while "Poor" and especially "Hazardous" are less represented, with only 20% and 10% respectively. This imbalance may lead to biased model performance, where minority classes like "Hazardous" are underpredicted. To address this, appropriate techniques such as resampling, applying class weights, or using evaluation metrics sensitive to imbalance (e.g., precision, recall, and F1-score) should be considered.

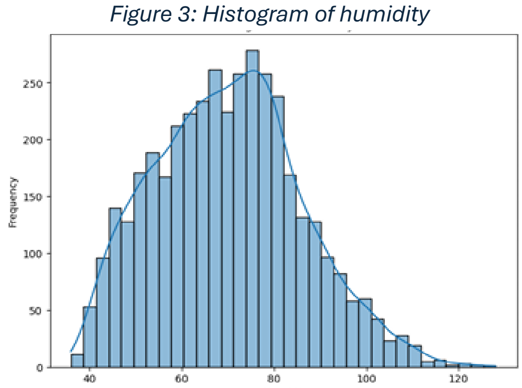
Distribution of other variables

Figure 2: Histogram of temperature



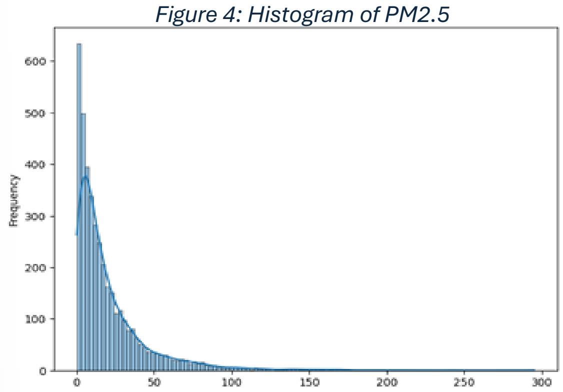
The distribution of temperature appears to be normal but slightly right skewed. Most of the temperature values are concentrated between 20°C and 35°C. There are fewer occurrences of temperatures below 15°C and above 45°C, indicating outliers or rare events.

Figure 3: Histogram of humidity



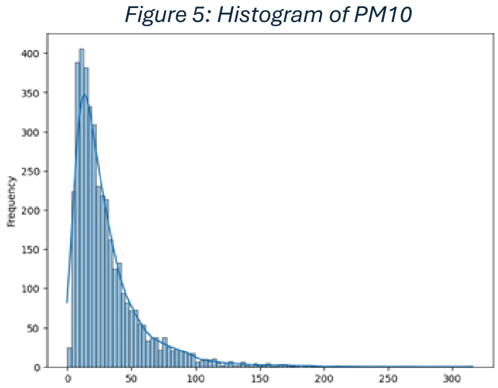
Humidity also shows a slight right-skewed distribution. The majority of the humidity values lie between 50% and 80%. Very high humidity values (above 100%) are relatively rare.

Figure 4: Histogram of PM2.5



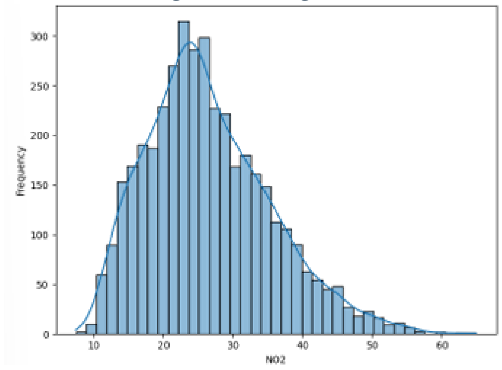
PM2.5 levels have a highly skewed distribution with a concentration of values at the lower range. Most PM2.5 values fall under 0 to 50 micrograms/m³, suggesting lower pollution levels for most cases. However, some extreme values above 100+ indicate sporadic periods of high pollution.

Figure 5: Histogram of PM10



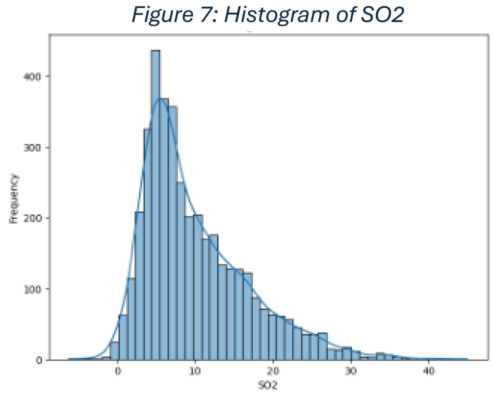
Like PM2.5, PM10 values show a right-skewed distribution. Most PM10 readings are concentrated between 0 and 50 micrograms/m³. A few values reach as high as 300, indicating rare but severe pollution episodes.

Figure 6: Histogram of NO2



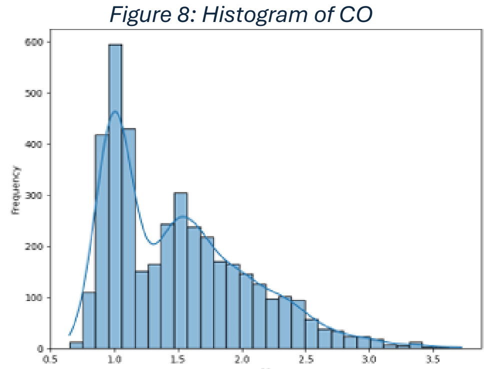
The NO2 levels follow a right-skewed distribution. The majority of NO2 values lie between 10 and 40 micrograms/m³. Very high levels (>50) are uncommon, indicating pollution spikes.

Figure 7: Histogram of SO2



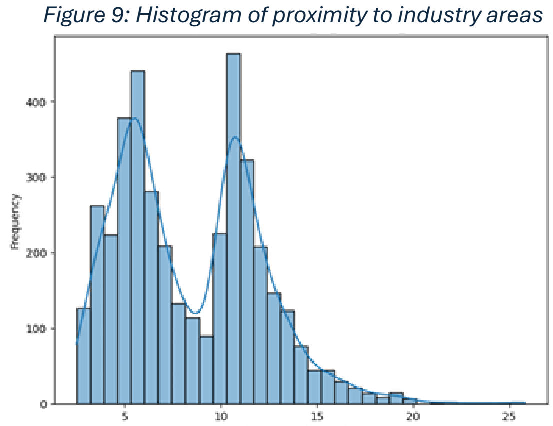
The SO2 levels also follow a right-skewed distribution. The majority of SO2 values lie between 0 and 10 micrograms/m³. Very high levels (>30) are uncommon, indicating pollution spikes.

Figure 8: Histogram of CO



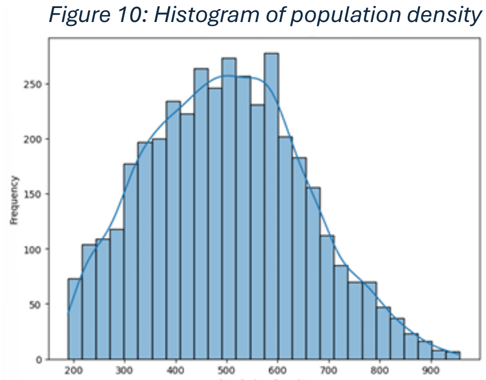
The distribution of CO is right-skewed. Most CO values are concentrated around 1.0 to 1.5. A smaller number of instances reach higher values, above 2.5, representing occasional spikes in CO levels. The skewness suggests occasional pollution events but a predominance of low CO levels.

Figure 9: Histogram of proximity to industry areas



The distribution of this variable appears to be bimodal (two peaks). One peak occurs around 5 units of proximity, and another peak is around 10 units. This bimodal behavior may indicate distinct clusters, possibly separating areas close and moderately far from industrial zones. Very few values exceed 15 units, suggesting that most data points represent locations relatively close to industrial areas.

Figure 10: Histogram of population density



Population density follows a normal-like distribution with slight skewness. The values are concentrated between 400 and 600, indicating typical population density for most regions in the dataset. The distribution tails off for lower densities (800), showing less frequent occurrences.

Bivariate analysis

Association between Air quality and Temperature

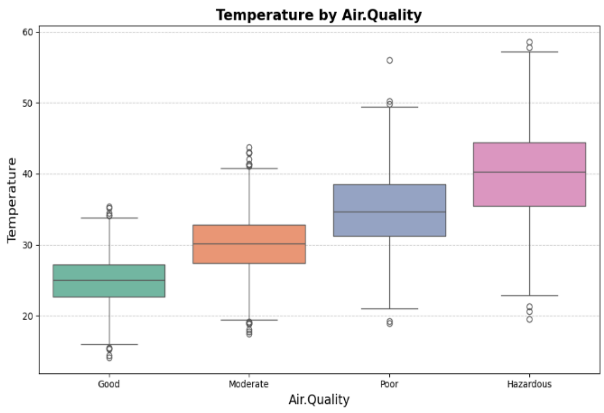


Figure 11: Boxplot of Temperature by Air quality

By Shapiro test, normality exists in temperature among air quality groups, but Levene's test confirms variance is not equal among groups of temperature. By boxplots medians of groups seem different and the highest is temperature of hazardous group. lowest temperature is temperature of good air quality group. By Kruskal Wallis and Dunn's test we confirmed that the median temperature of groups are different significantly. When air quality increases the median decreases. By Welch's Anova and Games-Howell tests all means are different significantly from each group's means.

Association between Air quality and Humidity

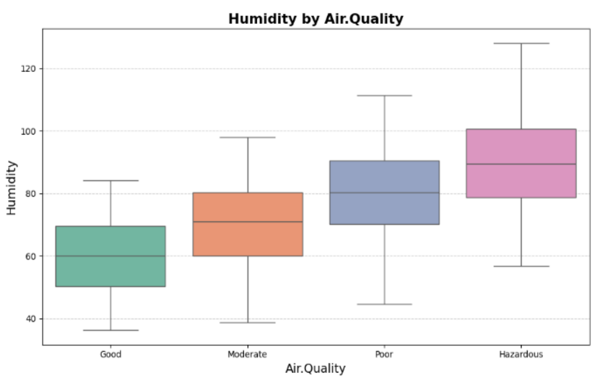


Figure 12: Boxplot of Humidity by Air Quality

The median humidity in different air quality groups are different significantly from each other according to Kruskal Wallis and Dunn's tests. When air quality increases the median of humidity decreases by boxplot. The Shapiro test confirms that normality assumption is violated among groups. There is no equal variance between different groups of humidity according to Levene's test. By Welch's Anova and games – Howell test mean of each group is different significantly from each other.

Association between Air quality and PM2.5

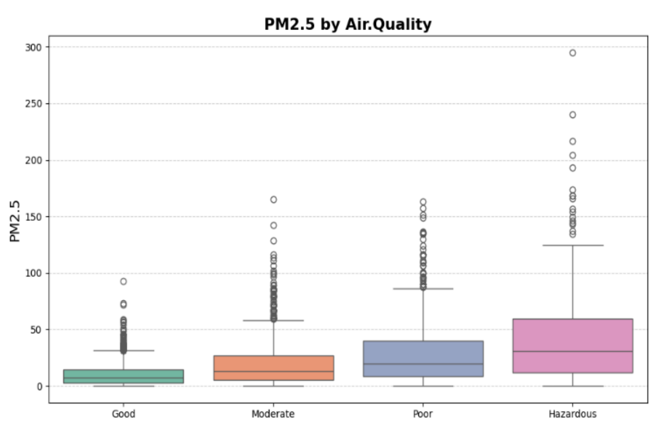


Figure 13: Boxplot of PM2.5 by Air Quality

The Shapiro test and Levene's test confirmed normality and equal variance among groups are violated. So, by Kruskal Wallis and Dunn's test medians of PM2.5 in each air quality group is different significantly from each other and Means of these groups are also di The Shapiro test and Levene's test confirmed normality and equal variance among groups are violated. So, by Kruskal Wallis and Dunn's

Association between Air quality and PM10

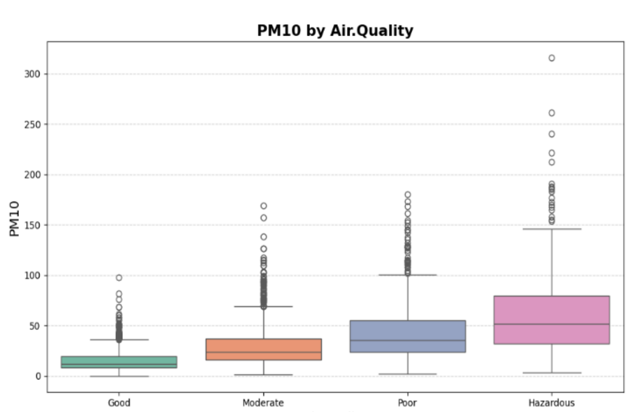


Figure 14: Boxplot of PM10 by Air Quality

The Shapiro test and Levene's test confirmed normality and equal variance among groups are violated. So, Kruskal Wallis and Dunn's test medians of PM10 in each air quality group are different significantly from each other and Means of these groups are also different significantly from each other according to Welch's Anova. By box plot we can see when air quality increases PM10 median increases.

Association between Air quality and NO2

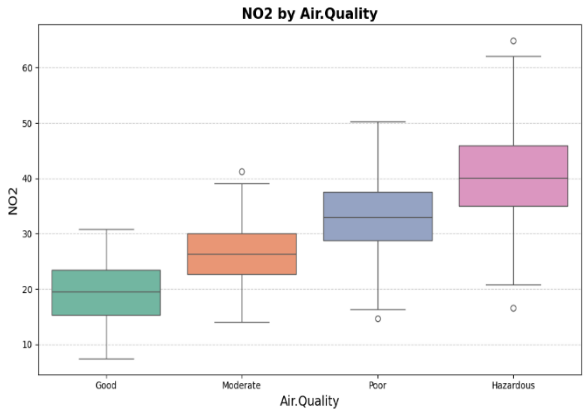


Figure 15: Boxplot of NO2 by Air Quality

By box plot we can see when air quality is increasing NO2 is increasing. By Shapiro test and Levene's test we confirm that normality and equal variance among groups are violated. According to Welch's Anova test and Games Howell test all means are different significantly from each other. All medians of groups differ significantly from each other by Kruskal Walli and Dunn's test. Lowest for good and highest for hazardous air quality group.

Association between Air quality and SO2

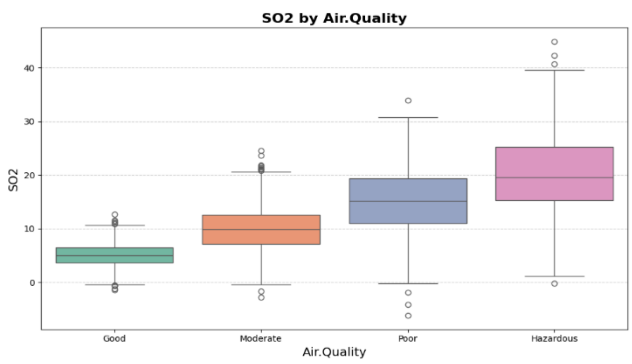


Figure 16: Boxplot of SO2 by Air Quality

By box plot, we can see that as air quality gets bad, SO2 levels increase. Shapiro test confirms normality among groups, but Levene's test shows unequal variance, so assumptions are violated. Welch's ANOVA and Games-Howell test show all means differ significantly from each other. Kruskal-Wallis and Dunn's test confirm all medians also differ significantly across groups, with lowest levels in good air quality and highest in hazardous air quality groups.

Association between Air quality and CO

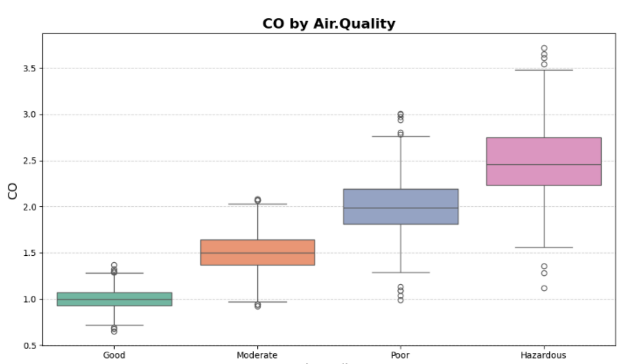


Figure 17: Boxplot of CO by Air Quality

By box plot, we can see that as air quality decreases, CO levels increase. Shapiro test confirms normality among groups, but Levene's test shows unequal variance assumptions are violated. Welch's ANOVA and Games-Howell test show all means differ significantly from each other. Kruskal-Wallis and Dunn's test confirm all medians also differ significantly across groups, with lowest levels in good air quality and highest in hazardous air quality groups.

Association between Air quality and Proximity to industry area

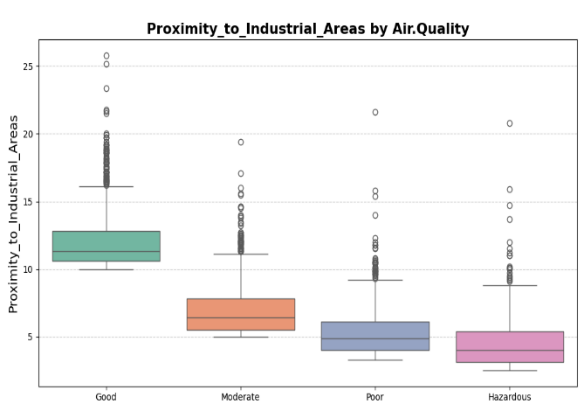


Figure 18: Boxplot of proximity to industry area by Air Quality

By box plot, we can see that as air quality deceases, proximity to industrial areas decreases. Shapiro test and Levene's test confirm that both normality and equal variance assumptions are violated among groups. Welch's ANOVA and Games-Howell test show all means differ significantly from each other. Kruskal-Wallis and Dunn's test confirm all medians also differ significantly across groups, with highest proximity in good air quality and lowest in hazardous air quality groups.

Association between Air quality and Population density

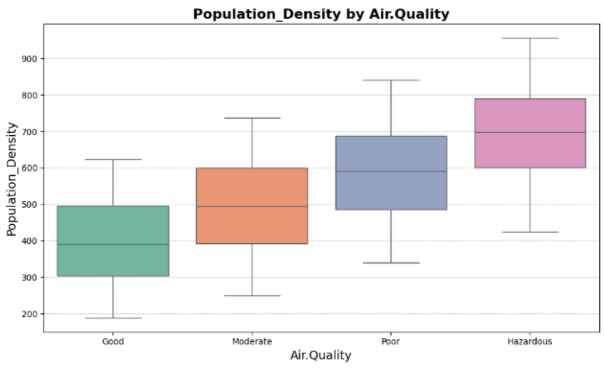
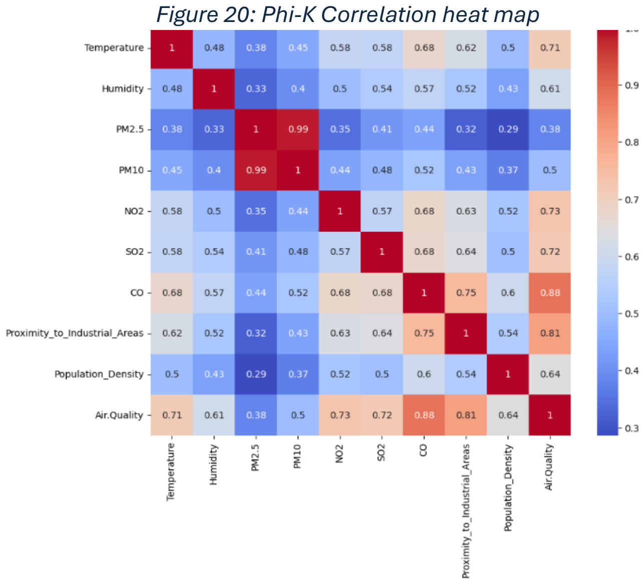


Figure 19: Boxplot of Population density by Air Quality

By box plot, we can see that as air quality decreases, population density increases. Shapiro test and Levene's test confirm that both normality and equal variance assumptions are violated among groups. Welch's ANOVA and Games-Howell test show all means differ significantly from each other. Kruskal-Wallis and Dunn's test confirm all medians also differ significantly across groups, with lowest density in good air quality and highest in hazardous air quality groups.

Correlation Analysis

Figure 20: Phi-K Correlation heat map



By correlation heatmap we can see clear patterns among the variables:

When one pollutant's level rises, others tend to increase as well.

PM2.5 and PM10 are nearly identical: These two particulate matter measures show an almost perfect correlation (r = 0.99), meaning they essentially provide the same information.

Air quality readings are strongly linked with:

- Carbon Monoxide (CO): 0.88
- Proximity to Industrial Areas: 0.81
- Nitrogen Dioxide (NO2): 0.73
- Sulfur Dioxide (SO2): 0.72

Temperature is moderately to strongly associated with several pollutants, especially CO (0.68) and both NO2 and SO2 (0.58 each).

Multicollinearity

To check for overlapping information among variables, a Variance Inflation Factor (VIF) analysis was performed:

Feature	VIF
Temperature	2.085547
Humidity	1.580398
PM2.5	28.706492
PM10	33.650673
NO2	2.253335
SO2	2.027349
CO	3.873047
Proximity_to_Industrial_Areas	2.281137
Population_Density	1.627177

Figure 21: Vif values

- Serious redundancy found: PM2.5 (VIF = 28.71) and PM10 (VIF = 33.65) have extremely high VIF values, confirming they are almost interchangeable.
- Temperature, Humidity, NO2, SO2, CO, Proximity to Industrial Areas, and Population Density all have VIFs below 4, indicating little to no redundancy.

Solving Redundancy with PCA

To address these overlaps, Principal Component Analysis (PCA) was used:

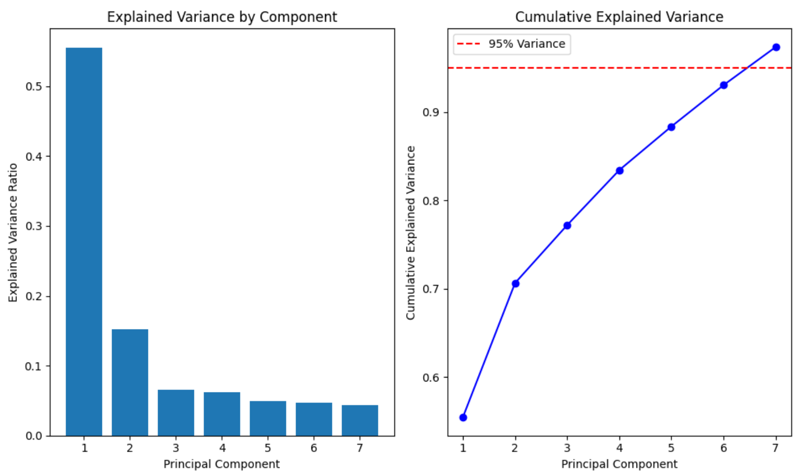


Figure 22: PCA plots

- Transformed the original 9 variables into 7 new, uncorrelated principal components. These 7 components together capture 97% of all the information (variance) in the original data. The first component alone explains 55% of the variance. The second adds another 15%. The remaining five each contribute between 5–7%.

Outlier analysis

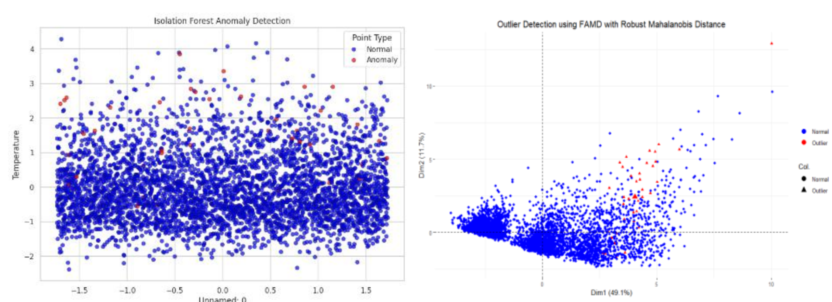


Figure 23: Outlier Detection plots from isolation forests

We used two complementary methods: Robust Mahalanobis Distance with FAMD and Isolation Forest. The Mahalanobis method identified multivariate deviations from core data structure, while Isolation Forest captured anomalous isolation patterns.

- Four extreme outliers in Relative Humidity (>100%) were removed as they violated physical constraints
- 40 additional outliers were consistently identified by both methods but retained to preserve data variability and maintain model generalizability

This approach balances data quality with information preservation, ensuring models train on both common and rare conditions.

Cluster Analysis

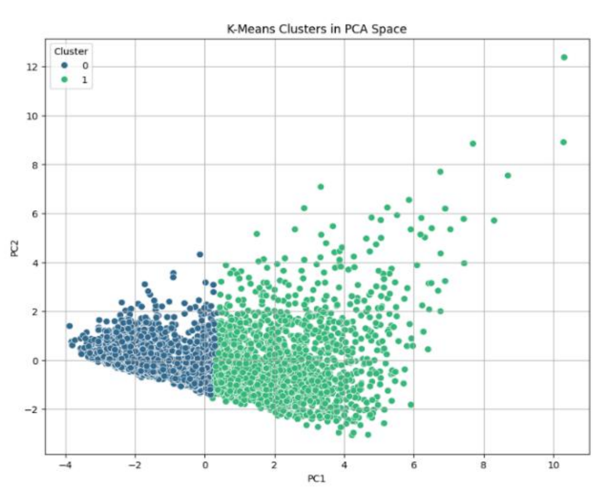


Figure 24: K-means Clusters plot in PCA space

- To get clear idea about clusters in this dataset we used k – means because of numerical variables. We did k means analysis for cluster n=2,3,4 and 5. Best silhouette score is 0.34 for 2 clusters.
- A silhouette score of 0.34 indicates that the dataset has some tendency to form two distinct groups, but boundaries between clusters are fuzzy.This supports the idea that the air quality data may not form clear separable groups independent of the labeled classes.

Suggestions for Quality Advance Analysis

Through the exploration data analysis we have identified the existence of outliers, multi-collinearity. Thus, the following models were suggested:

- **Logistic Regression** - Handles multi-collinearity well through regularization techniques. Reduces dimensionality impact and outlier effects are minimized because it penalizes large coefficients.
- **Random Forest** - Naturally handles multi-collinearity and redundant features through ensemble learning. Robust to outliers due to bootstrap sampling and majority voting mechanisms.
- **XG Boost** - Advanced gradient boosting algorithm that is more complex than Random Forest. Handles feature interactions effectively and provide built-in regularization to address multi-collinearity.
- **Support Vector Machine (SVM)** - Effective for high-dimensional data and naturally handles multi-collinearity through kernel transformations. Less sensitive to outliers, especially when using appropriate kernel functions and regularization parameters.

Appendix and technical details

- Colab code
- https://colab.research.google.com/drive/1BB0_EW1FTAUVlLs2OiXXyMZAZTc7KYHo?usp=sharing

References

- https://byjus.com/biology/air-pollution-control/ - Air Pollution Control – Causes, Effects, and Prevention.
- https://www.youtube.com/watch?v=Tds3k97aAzo - What is air pollution? [Video]. YouTube.
- https://manifold.open.umn.edu/read/chapter-6-steps-for-bivariate-analysis-and-results/section/e941d76f-63c4-48f2-9163-23539d4e9206.
- https://www.geeksforgeeks.org/data-visualization/box-plot-in-python-using-matplotlib/
- https://phik.readthedocs.io/en/latest
- https://www.geeksforgeeks.org/data-analysis/principal-component-analysis-pca/